

ขับเคลื่อน Observability ยุคใหม่ด้วย eBPF: DeepFlow

"เราเชื่อว่าการเรียนรู้ทำให้ชีวิตคุณดีขึ้น"

-Jumpbox Team-



Jumpbox®

Tech Passion | Professional | Sharing



JoJo
(Cloud Native Stylist)



CLOUD NATIVE BANGKOK COMMUNITY GROUP

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Co-Founder
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CEO
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Head of Platform
Engineering
Ascend Money

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Damrongsak Reetanon
Chief Information Security
Officer

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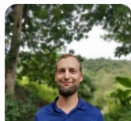
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Head of Platform
Engineering
Ascend Bit Company (ABC)

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 Chaos Mesh CNCF INCUBATING	 Litmus CNCF INCUBATING
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 chaosk8s	 steadybit
 ChaosMeta	 K8sE2E
 krkn	

Feature Flagging

Open Feature

CNCF INCUBATING

FeatHub

Bucket

Bucketeer

DevCycle

Flagsmith

Flippi

LaunchDarkly

Split

Unleash

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Observability (June 2017)

<https://github.com/deepflowio/deepflow>

[README](#) [Code of conduct](#) [Apache-2.0 license](#)

DeepFlow[®]

Instant Observability for Cloud & AI Applications

Zero Code, Full Stack, eBPF & Wasm

[DOI: 10.5281/zenodo.18577772](#) [release](#) [invalid](#) [G-Star](#) [20 Stars](#) [docker pulls](#) [license](#) [Apache-2.0](#)

[English](#) | [简体中文](#) | [日本語](#)

What is DeepFlow

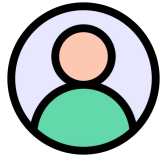
The DeepFlow open-source project aims to provide deep observability for complex cloud-native and AI applications. DeepFlow implemented **Zero Code** data collection with eBPF for metrics, distributed tracing, request logs and function profiling, and is further integrated with **SmartEncoding** to achieve **Full Stack** correlation and efficient access to all observability data. With DeepFlow, cloud-native and AI applications automatically gain deep observability, removing the heavy burden of developers continually instrumenting code and providing monitoring and diagnostic capabilities covering everything from code to infrastructure for DevOps/SRE teams.

Key Features

- **Universal Map for Any Service:** DeepFlow provides a universal map with **Zero Code** by eBPF for production environments, including application services, AI services, and infrastructure services in any language. In addition to analyzing common protocols, Wasm plugins are supported for your private protocols. **Full Stack** golden signals of applications and infrastructures are calculated, pinpointing performance bottlenecks at ease.

Let's getting started with

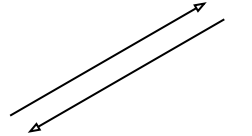
Microservices Story (Ecommerce)



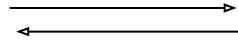
User



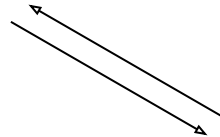
Dashboard



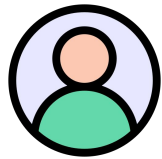
Microservice



Microservice



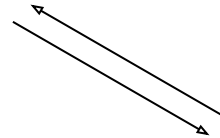
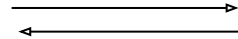
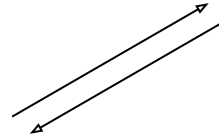
Microservice



User



Dashboard



User

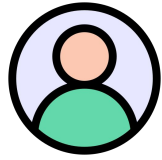
Microservice

Wallet

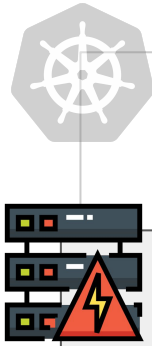
Microservice

Promotion

Microservice



User



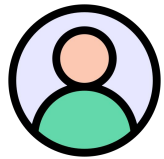
**High
Load, Latency, CPU, Mem, ...**

User

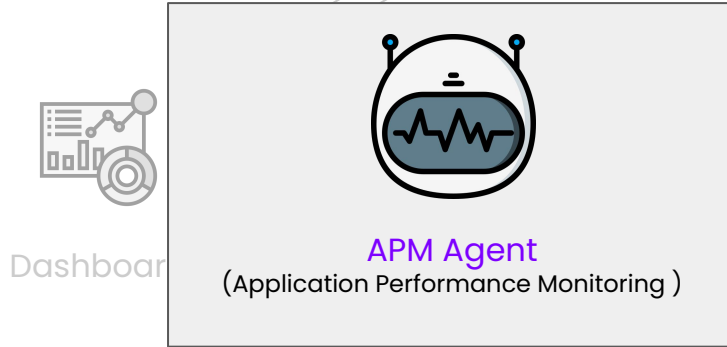
Promotion

Microservice

Observability
comes in



User



Dashboard

User

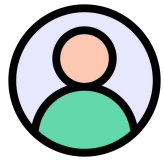
Microservice

llet

ervice

Promotion

Microservice



User



Dashboard



User

Microservice



Wallet

Microservice



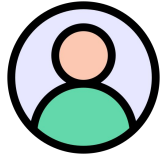
Promotion

Microservice

It's coming
with **complex**

Solution?

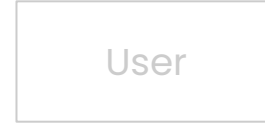




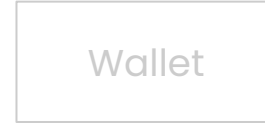
User



Dashboard



Microservice



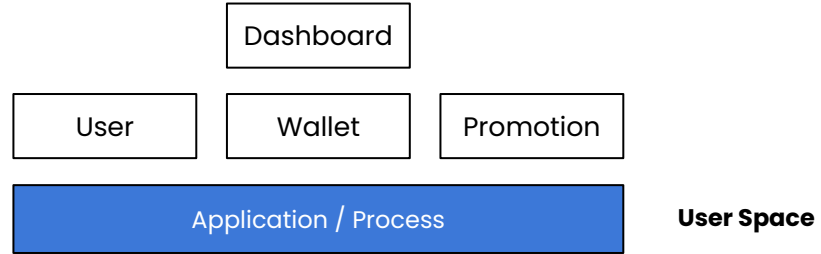
Microservice



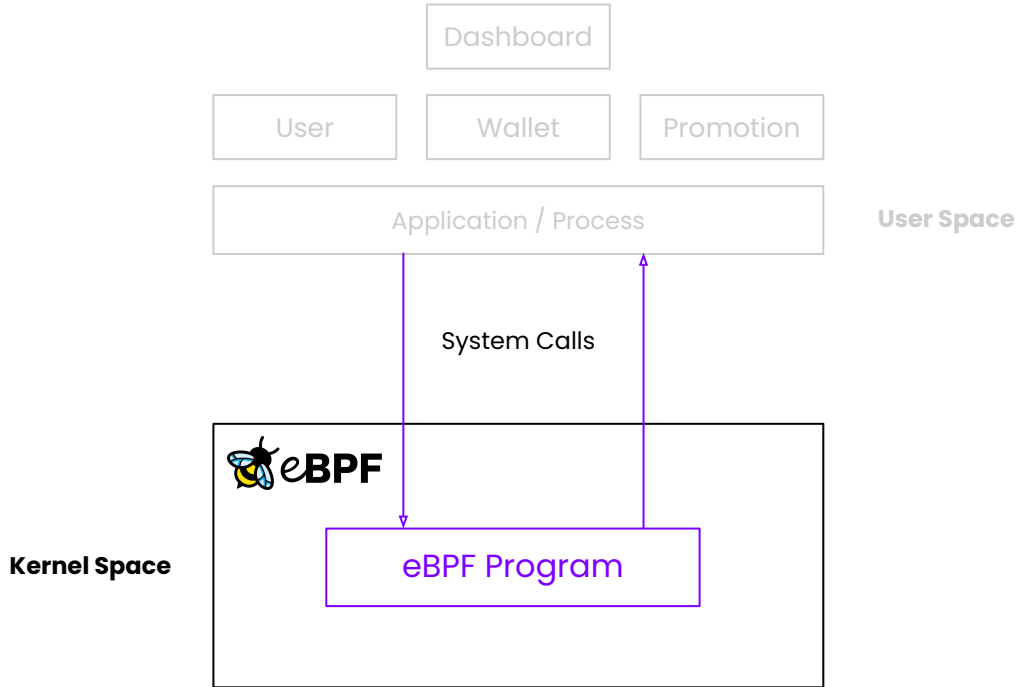
Microservice



eBPF with logical structure (Cont.)



eBPF with logical structure (Cont.)

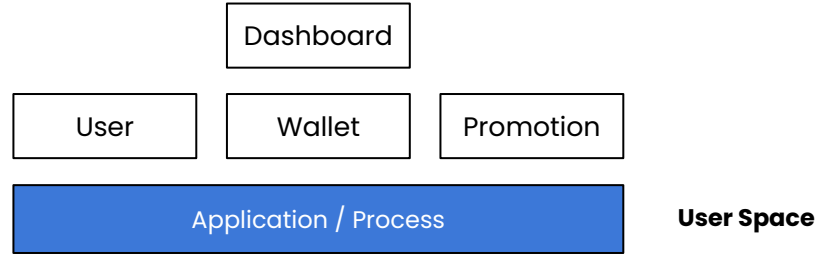


ทำให้เห็นภาพขึ้น

Example in Real Life:
Application use System Calls

Application:
ls, curl, cd, ...etc

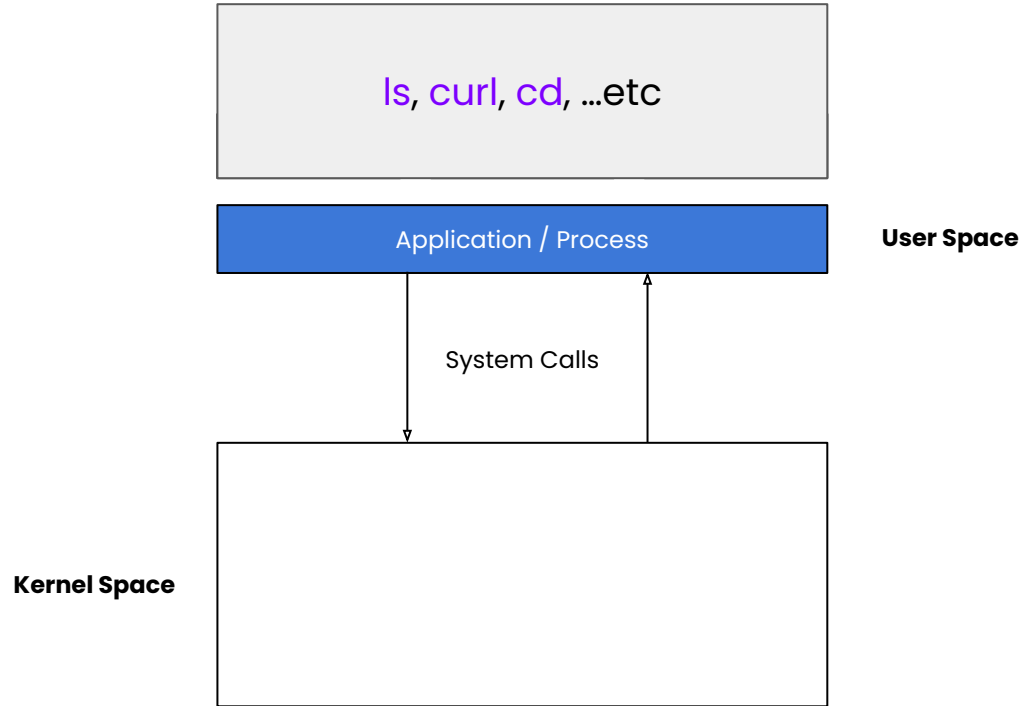
Demo System Calls (Cont.)



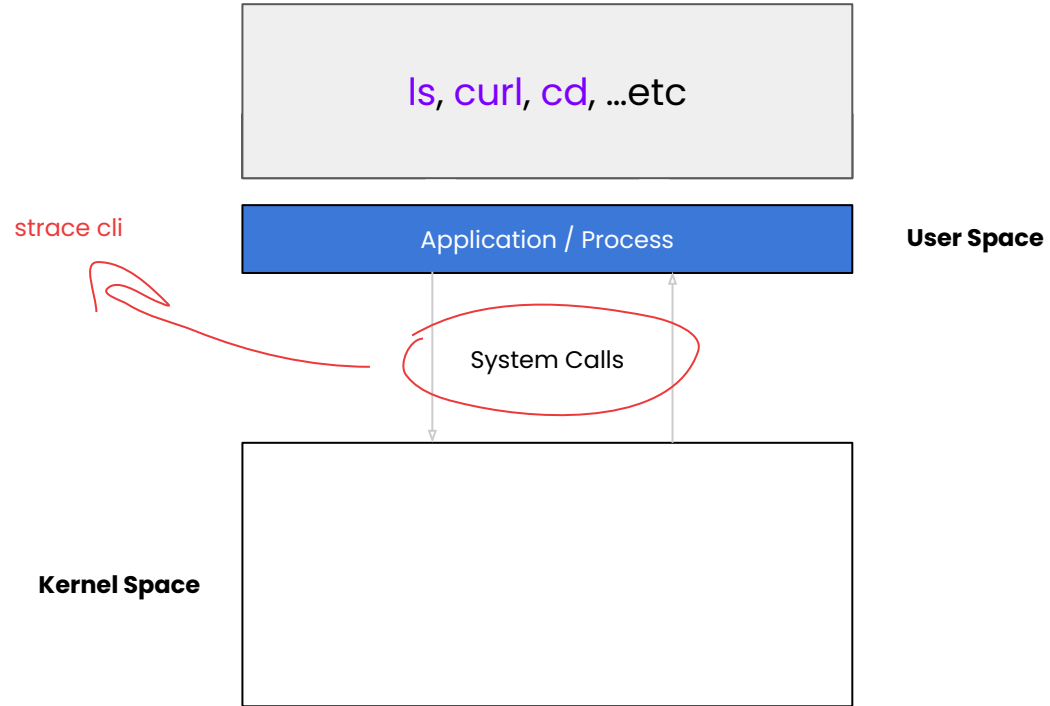
Demo System Calls (Cont.)



Demo System Calls (Cont.)



Demo System Calls (Cont.)



eBPF with logical structure (Cont.)

```
sikiry!@lima-ubuntu:/Users/sikiry!$ strace ls
execve("/usr/bin/ls", ["ls"], 0xffffd7eddf50 /* 24 vars */) = 0
brk(NULL)                                = 0xb7280560d000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0xf19d2ab9a000
faccessat(AT_FDCWD, "/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=21041, ...}) = 0
mmap(NULL, 21041, PROT_READ, MAP_PRIVATE, 3, 0) = 0xf19d2ab94000
close(3)                                 = 0
openat(AT_FDCWD, "/lib/aarch64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0\267\0\1\0\0\0\0\0\0\0\0\0\0"... , 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=264568, ...}) = 0
mmap(NULL, 402992, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_DENYWRITE, -1, 0) = 0xf19d2ab01000
mmap(0xf19d2ab10000, 337456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0) =
    0xf19d2ab10000
munmap(0xf19d2ab01000, 61440)            = 0
munmap(0xf19d2ab63000, 1584)            = 0
mprotect(0xf19d2ab44000, 110592, PROT_NONE) = 0
```

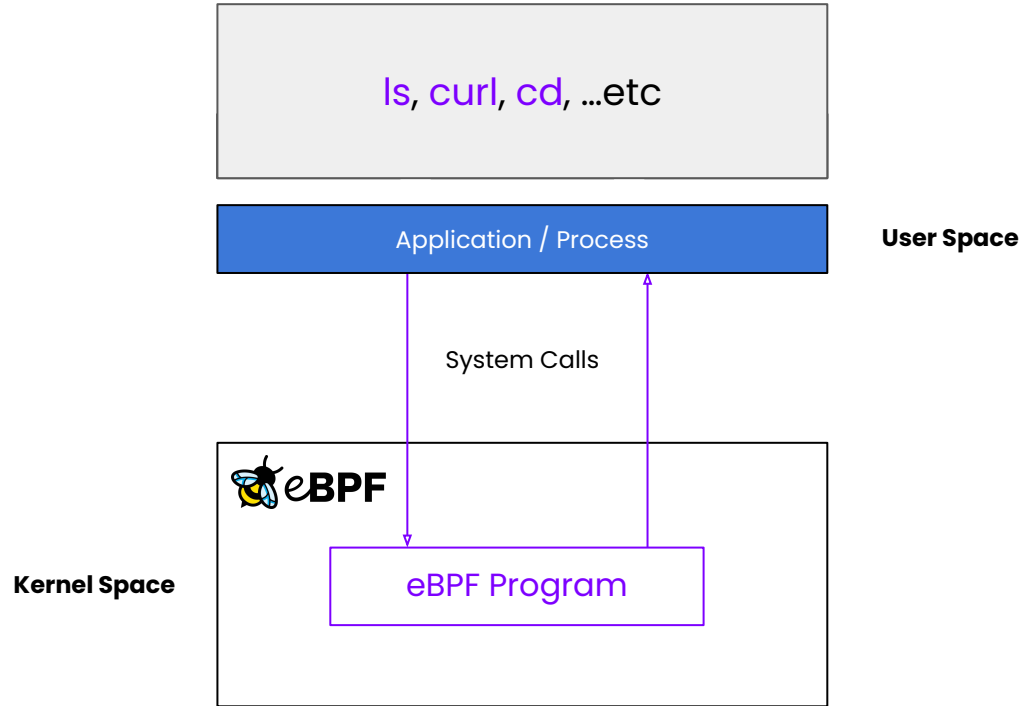
eBPF with logical structure (Cont.)

หัวใจสำคัญ (ขอเก็บไว้ก่อน)

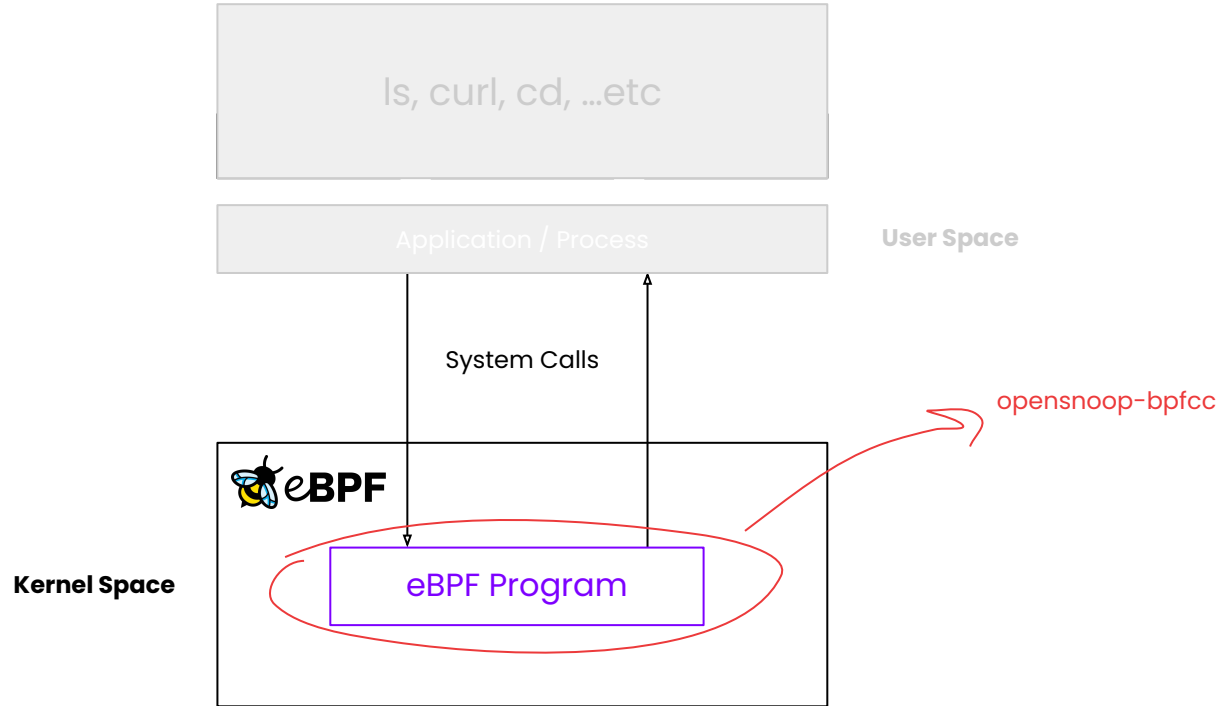
```
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fstat(3, {st_mode=S_IFREG|0644, st_size=264568, ...}) = 0  
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```

กลับมาที่ eBPF

eBPF with logical structure (Cont.)



eBPF with logical structure (Cont.)



eBPF with logical structure (Cont.)

```
sikiryl@lima-ubuntu: /Users/sikiryl
sikiryl@lima-ubuntu: /Users/sikiryl (ssh)
sikiryl@lima-ubuntu: /Users/sikiryl$ sudo opensnoop-bpfcc
PID    COMM          FD ERR PATH
2815   ls             3  0  /etc/ld.so.cache
2815   ls             3  0  /lib/aarch64-linux-gnu/libselinux.so
.1
2815   ls             3  0  /lib/aarch64-linux-gnu/libgcc_s.so.1
2815   ls             3  0  /lib/aarch64-linux-gnu/libm.so.6
2815   ls             3  0  /lib/aarch64-linux-gnu/libc.so.6
2815   ls             3  0  /lib/aarch64-linux-gnu/libpcr2-8.so
.0
2815   ls             3  0  /proc/filesystems
2815   ls             3  0  /proc/self/maps
2815   ls            -1 20  /usr/lib/cargo/bin/coreutils/ls/en-U
S.ftl
2815   ls            -1 20  /usr/lib/cargo/bin/coreutils/ls/en-U
S.ftl
2815   ls             3  0  .

sikiryl@lima-ubuntu: /Users/sikiryl/apisix (ssh)
sikiryl@lima-ubuntu: /Users/sikiryl/
apisix$ ls
apisixroute.yaml
ingress-controller.yaml
values.yaml
sikiryl@lima-ubuntu: /Users/sikiryl/
apisix$
```





Website

Submit

Reference(s):

- [Keynote: eBPF](#)



Website

```
function hello() {  
  alert('hello')  
}  
</script>  
<form onsubmit="return false;">  
  <input type="submit"  
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</form>
```

Reference(s):

- [Keynote: eBPF](#)



Website

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</> Process

execve()

Operating
System



System Calls



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Website

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</> Process

execve()

Operating
System



System Calls

```
int syscall__ret_execve(struct pt_regs *ctx)  
{  
    struct comm_event event = {  
        .pid = bpf_get_current_pid_tgid() >> 32,  
        .type = TYPE_RETURN,  
    };  
    bpf_get_current_comm(&event.comm, sizeof(event.comm));  
    comm_events.perf_submit(ctx, &event, sizeof(event));  
    return 0;  
}
```



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Website

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Reference(s):
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execve()

Operating
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System Calls

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}
```



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execve = หัวใจสำคัญ

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exe = execute = สั่งรัน program

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exe	= execute	= สั่งรัน program
v	= vector	= รับค่า array arguments

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Reference(s):

- [Keynote: eBPF](#)



</> Process

execve()

Operating
System



System Calls

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    comm_events.perf_submit(ctx, &event, sizeof(event));  
    return 0;  
}
```



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How does eBPF work?

 **eBPF** Language



 **eBPF** Runtime

Make it Simple

 eBPF Language



 eBPF Runtime

Runtime

Runtime

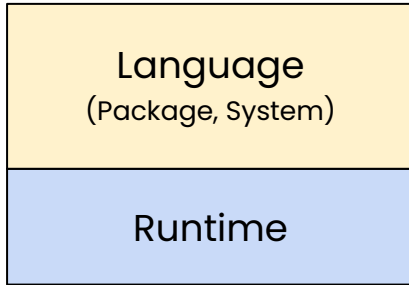
 nodejs

OpenJDK

 .NET

 GO

 Rust



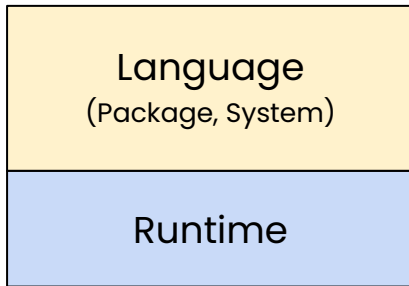
 nodejs

OpenJDK

 .NET

 GO

 Rust



*.js

node nodejs



*.jar

OpenJDK



*.dll,
*.exe

.NET

*.go

GO

*.rs

Rust



Language
(Package, System)

Runtime



*.js

node nodejs



*.jar

OpenJDK



*.dll,
*.exe

.NET

*.go

GO

*.rs

Rust

Accurate Definition

" **eBPF** is a programming language & runtime to extend operating system"

-Thomas Graf, CTO, Isovalent -



Reference(s):

- [Keynote: eBPF](#)
- [KubeCon2022 eBPF Day North America](#)

Practical Comparison

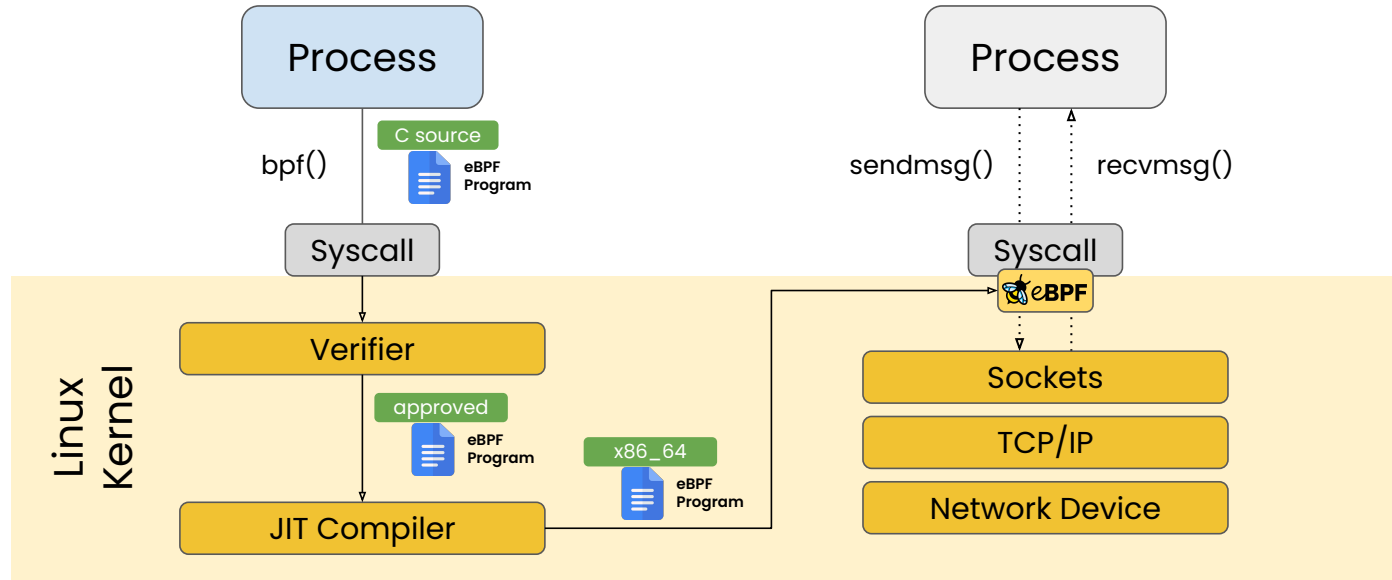
" **eBPF** is like JavaScript/Lua but for Kernel Developers"

–Thomas Graf, CTO, Isovalent –



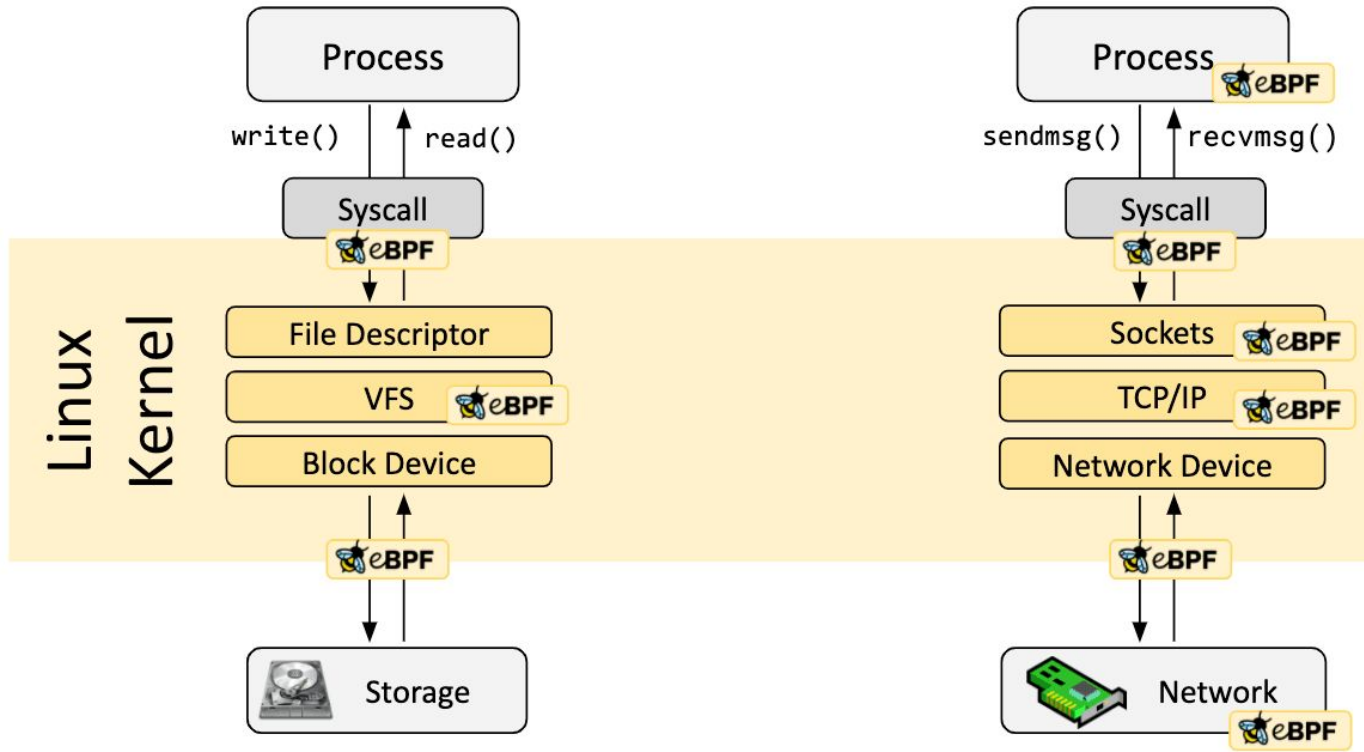
Reference(s):

- [Keynote: eBPF](#)
- [KubeCon2022 eBPF Day North America](#)



The runtime accepts bytecode, verifies it, just-in-time compiles it, and runs it at the requested hook point.

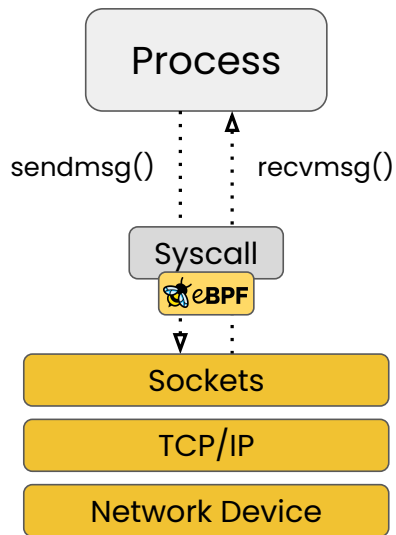
Reference(s):
- [What is ebpf](#)



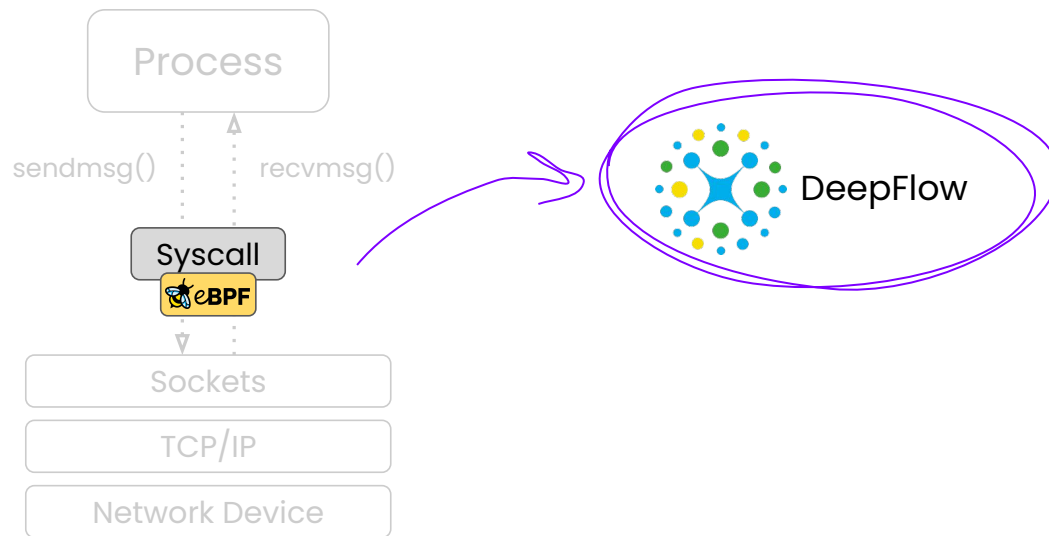
Reference(s):
- [What is ebpf](#)

In observability area **we focus**

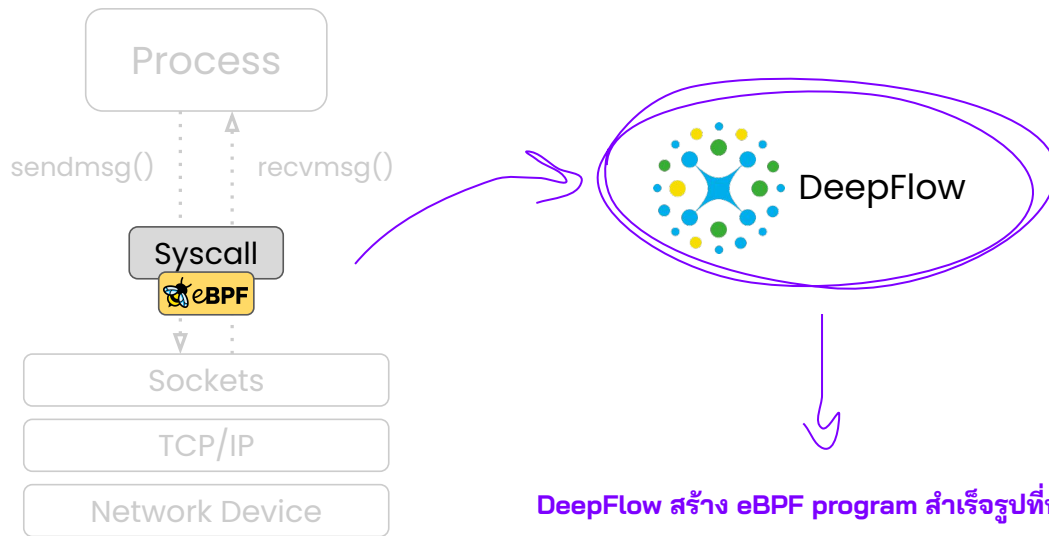
Why DeepFlow (Cont.)



Why DeepFlow (Cont.)



Why DeepFlow (Cont.)

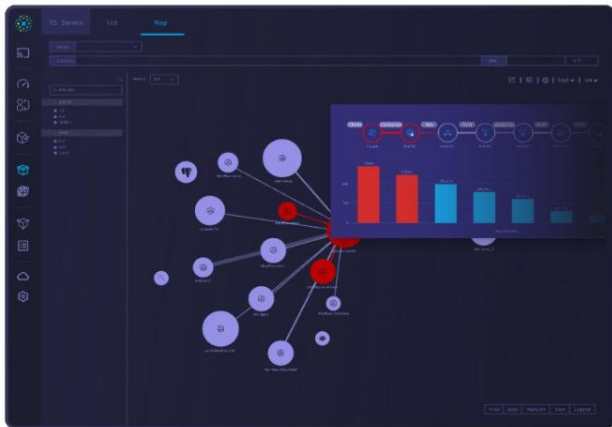


DeepFlow สร้าง eBPF program สำเร็จรูปที่ทำ Filter ให้
เข้าใจง่ายและใช้ได้จริง

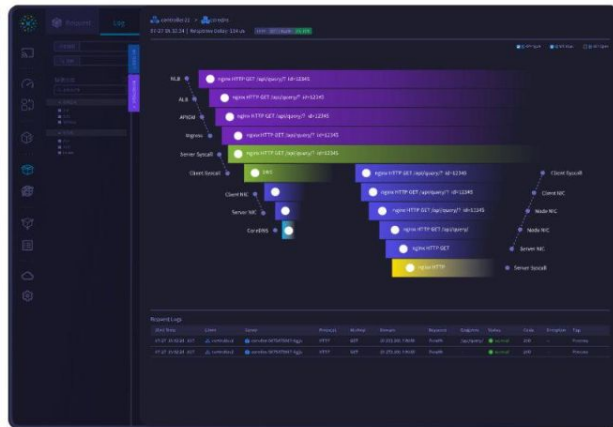
เช่น

- Zero-Code Observability
- Universal Service Map (Golden, RED)
- Distributed Tracing
- Support Multiple Protocols (gRPC, HTTP, ...)

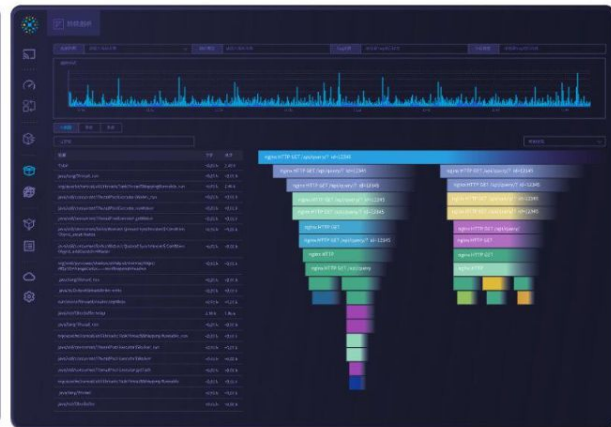
Why DeepFlow (Cont.)



Universal Map for Any Service

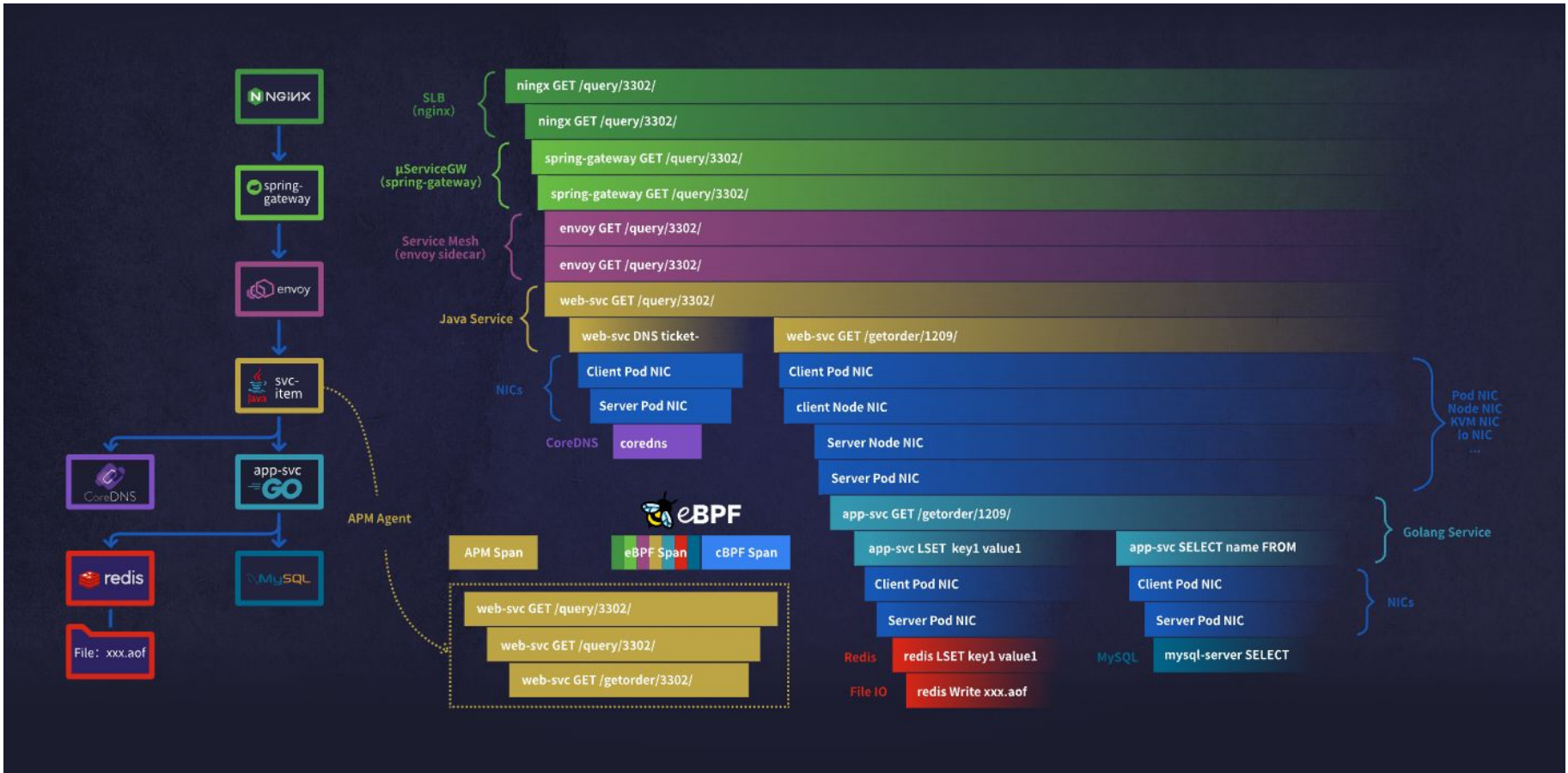


Distributed Tracing for Any Request



Continuous Profiling for Any Function

Reference(s):
- [DeepFlow edition](#)



Reference(s):
 - [eBPF: The Key Technology to Observability](#)

NIC = Network Interface Card

Why DeepFlow: Community Edition(CE) (Cont.)

Universal Map (AutoMetrics)

- Automatically collects application, network, and system full-stack performance metrics based on eBPF/cBPF
- Automatically collects TCP/UDP flow logs based on eBPF/cBPF
- Automatically collects application call logs for HTTP1/2, HTTPS (Golang/openssl), Dubbo, gRPC, SOFARPC, FastCGI, bRPC, MySQL, PostgreSQL, Redis, MongoDB, Kafka, MQTT, AMQP (RabbitMQ), OpenWire (ActiveMQ), NATS, Pulsar, ZMQ (ZeroMQ), RocketMQ, DNS, etc., based on eBPF/cBPF

Reference(s):

- [DeepFlow edition](#)

Why DeepFlow: Community Edition(CE) (Cont.)

Distributed Tracing (AutoTracing)

- Automatically traces microservice distributed call chains based on eBPF/cBPF

Continuous Profiling (AutoProfiling)

- Supports zero-intrusion On-CPU continuous profiling based on eBPF
- Supports JVM languages, compiled languages like C/C++/Golang/Rust

Reference(s):

- [DeepFlow edition](#)

Why DeepFlow: Community Edition(CE) (Cont.)

Integration

- Integrates Prometheus/Telegraf metrics data to solve data silos and high cardinality issues
- Integrates OpenTelemetry/SkyWalking tracing data to achieve full-stack distributed tracing
- Integrates Vector log data to solve data silos and resource consumption issues

Reference(s):

- [DeepFlow edition](#)

Why DeepFlow: Community Edition(CE) (Cont.)

AutoTagging

- Supports synchronization of public cloud resource tags and automatically injects them into all observability data
- Supports synchronization of container resource tags and custom labels, and automatically injects them into all observability data
- Supports SmartEncoding for high-performance data tag storage

Reference(s):

- [DeepFlow edition](#)

Why DeepFlow: Community Edition(CE) (Cont.)

Integration and Management

- Supports using Grafana to display metrics and tracing data
- Supports unified monitoring of multiple K8s clusters and regular cloud servers
- The collection agent supports running on K8s nodes, Serverless Pods, and Linux Host environments
- Supports deployment on X86 and ARM architectures

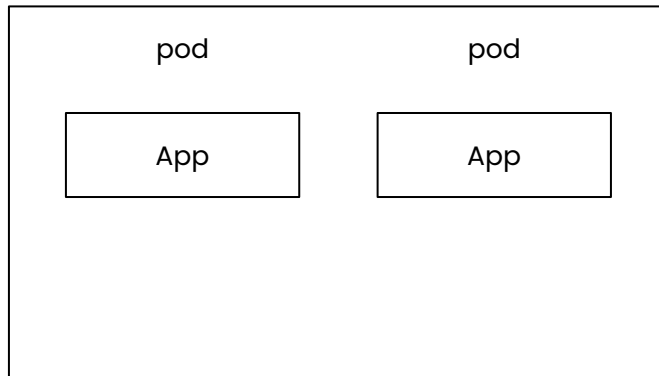
Reference(s):

- [DeepFlow edition](#)

Monitoring: Traditional vs eBPF

Monitoring: Traditional vs eBPF(Cont.)

Traditional

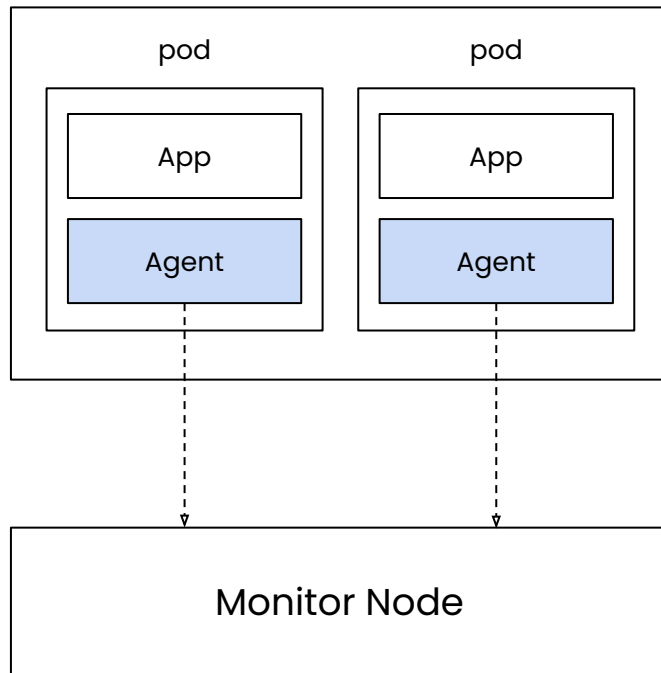


Reference(s):

- [DeepFlow Architecture](#)

Monitoring: Traditional vs eBPF(Cont.)

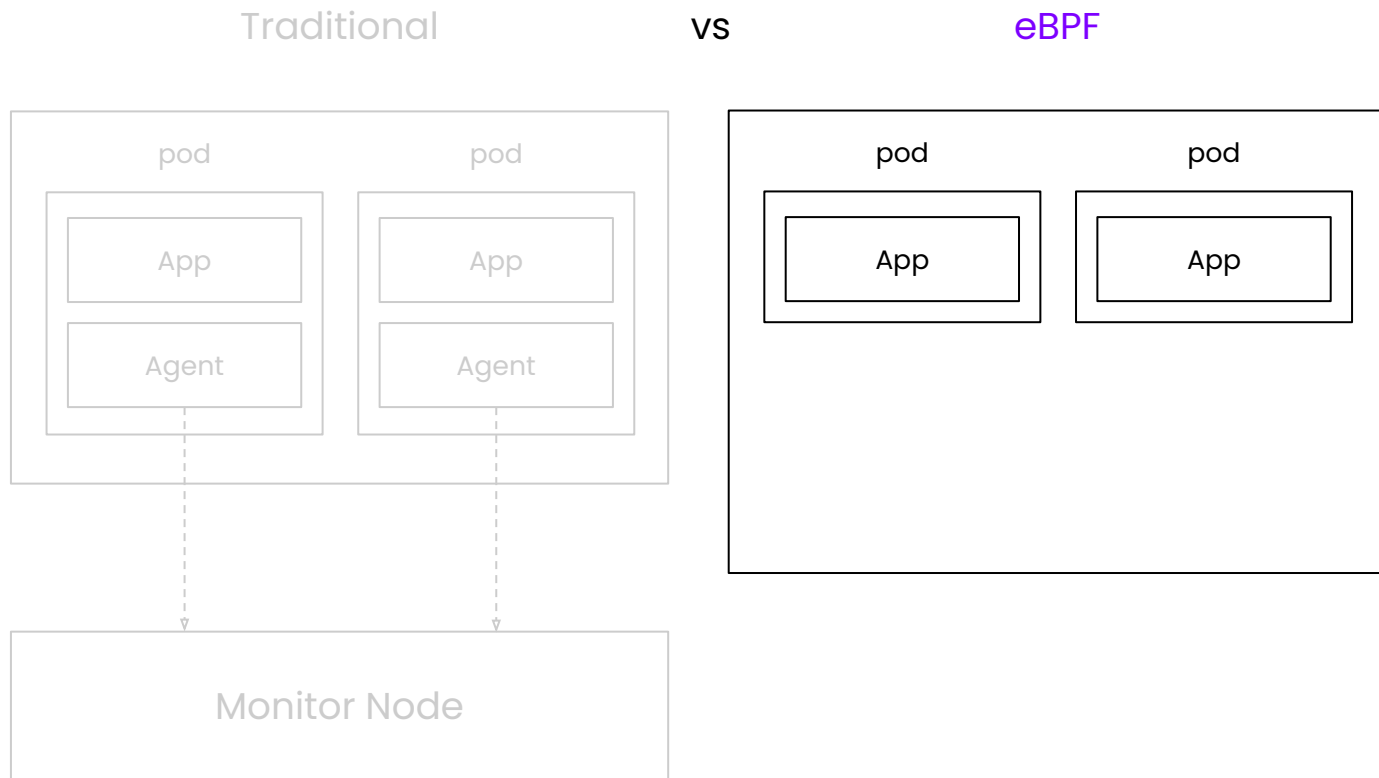
Traditional



Reference(s):

- [DeepFlow Architecture](#)

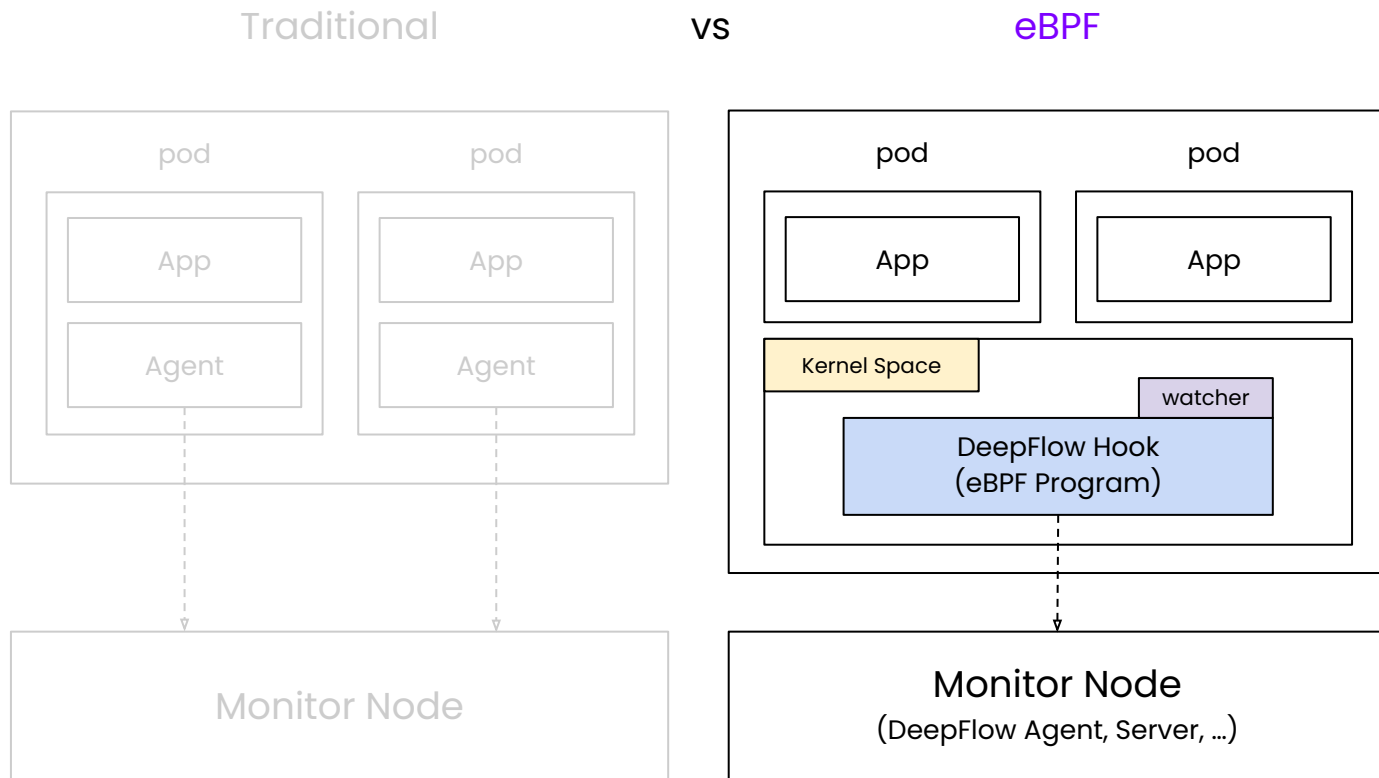
Monitoring: Traditional vs eBPF(Cont.)



Reference(s):

- [DeepFlow Architecture](#)

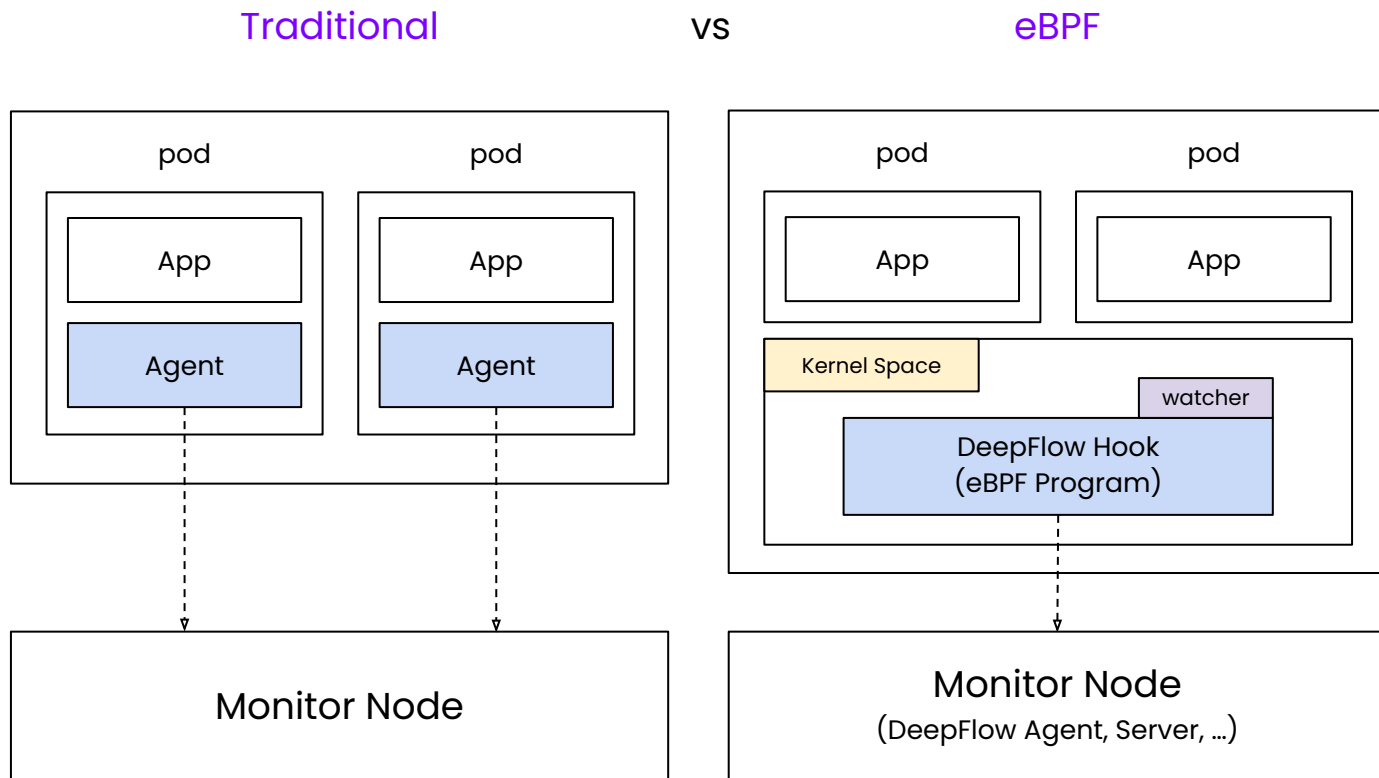
Monitoring: Traditional vs eBPF(Cont.)



Reference(s):

- [DeepFlow Architecture](#)

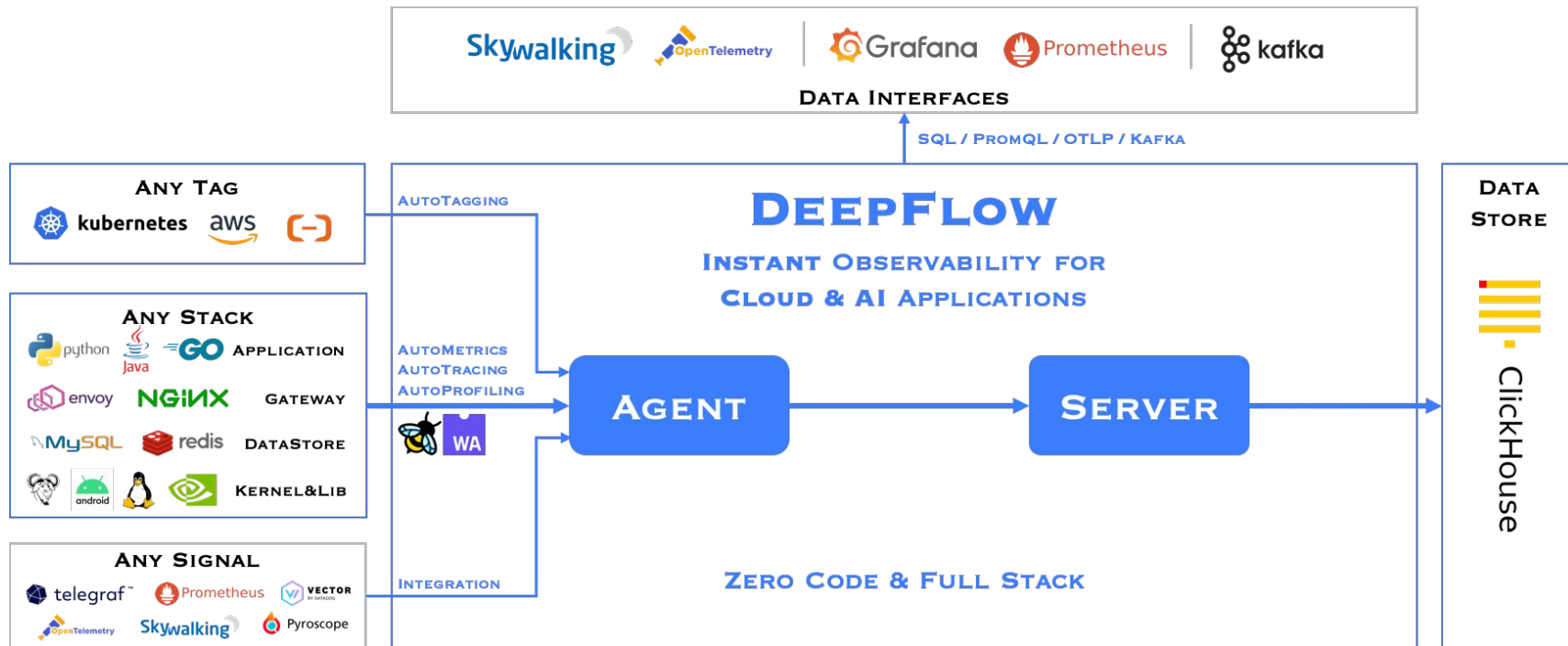
Monitoring: Traditional vs eBPF(Cont.)



Reference(s):

- [DeepFlow Architecture](#)

DeepFlow Architecture(Cont.)



Reference(s):

- [DeepFlow Architecture](#)



Demo

DeepFlow Demo (Cont.)

Prerequisite Command:

```
$ helm repo add deepflow https://deepflowio.github.io/deepflow
$ helm repo update deepflow
```

Install Command:

```
$ helm install deepflow \
  --namespace deepflow --create-namespace \
  -f values-custom.yaml \
  deepflow/deepflow
```

values-custom.yaml:

```
global:
  allInOneLocalStorage: true
```

Reference(s):

- [Install Command](#)

DeepFlow Demo (Cont.)

```
Context: colima [RW]
Cluster: colima
User: colima
K9s Rev: v0.50.18
K8s Rev: v1.30.2+k3s1
CPU: 9%
MEM: 56%
```

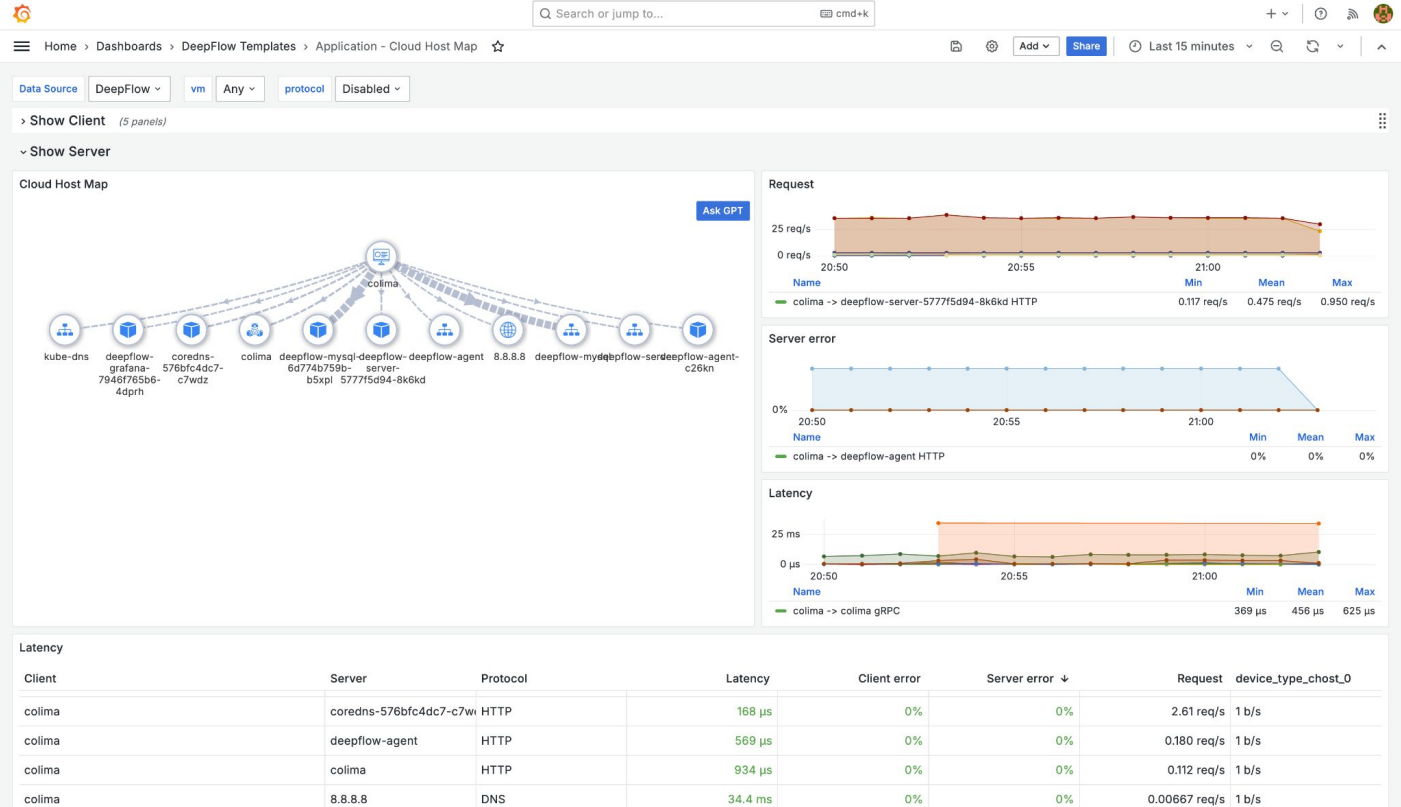
NAME	PF	READY	STATUS	RESTARTS	CPU	%CPU/R	%CPU/L	MEM	%MEM/R	%MEM/L	IP	NODE
deepflow-agent-c26kn	●	1/1	Running	0	18	18	1	444	347	57	10.42.0.38	colima
deepflow-app-6fd47db4d7-zld2z	●	1/1	Running	0	17	n/a	n/a	676	n/a	n/a	10.42.0.34	colima
deepflow-clickhouse-0	●	1/1	Running	0	35	n/a	n/a	1801	n/a	n/a	10.42.0.35	colima
deepflow-grafana-7946f765b6-4dprh	Ⓢ	1/1	Running	2	2	n/a	n/a	160	n/a	n/a	10.42.0.37	colima
deepflow-mysql-6d774b759b-b5xpl	●	1/1	Running	0	6	n/a	n/a	625	n/a	n/a	10.42.0.36	colima
deepflow-server-5777f5d94-8k6kd	●	1/1	Running	0	23	n/a	n/a	809	n/a	n/a	10.42.0.33	colima

<pod>

Reference(s):

- [Install Command](#)

DeepFlow Demo (Cont.)



Reference(s):
- [Install Command](#)

DeepFlow Demo (Cont.)

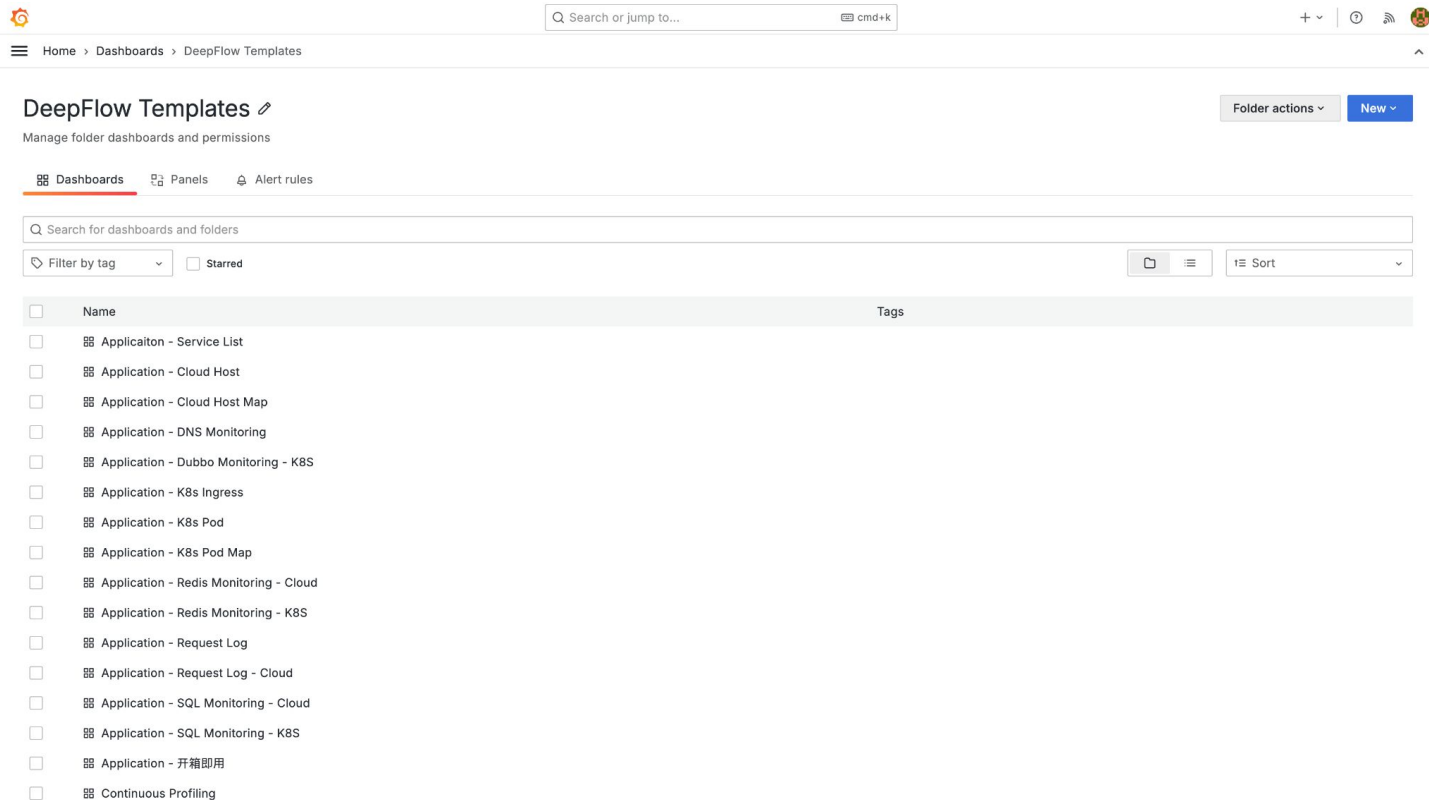
The screenshot shows the DeepFlow System dashboard. At the top, there's a navigation bar with a search bar, a command prompt icon (cmd+k), and user profile icons. Below the navigation bar, the breadcrumb path is "Home > Dashboards > DeepFlow System". The main heading is "DeepFlow System" with an edit icon. To the right of the heading are "Folder actions" and a "New" button. Below the heading is the subtitle "Manage folder dashboards and permissions". There are three tabs: "Dashboards" (active), "Panels", and "Alert rules". A search bar for "Search for dashboards and folders" is present. Below the search bar are filters for "Filter by tag" and "Starred". To the right of the filters are icons for folder view and list view, and a "Sort" dropdown. The main content area is a table with a header row containing "Name" and "Tags". The table lists various dashboards, each with a checkbox on the left and a gear icon on the right. The list includes: Capacity Estimation, ClickHouse Cluster Analysis, ClickHouse Data Analysis, ClickHouse Query Analysis, ClickHouse Tags, DeepFlow Agent, DeepFlow Alert Analysis, DeepFlow Controller - Prometheus, DeepFlow Resource, DeepFlow Server, DeepFlow Server - Controller, DeepFlow Server - Ingester, DeepFlow Server - Ingester - Prometheus, DeepFlow Server - Querier, and MySQL Tables.

☐	Name	Tags
☐	Capacity Estimation	
☐	ClickHouse Cluster Analysis	
☐	ClickHouse Data Analysis	
☐	ClickHouse Query Analysis	
☐	ClickHouse Tags	
☐	DeepFlow Agent	
☐	DeepFlow Alert Analysis	
☐	DeepFlow Controller - Prometheus	
☐	DeepFlow Resource	
☐	DeepFlow Server	
☐	DeepFlow Server - Controller	
☐	DeepFlow Server - Ingester	
☐	DeepFlow Server - Ingester - Prometheus	
☐	DeepFlow Server - Querier	
☐	MySQL Tables	

Reference(s):

- [Install Command](#)

DeepFlow Demo (Cont.)



The screenshot shows the 'DeepFlow Templates' page in the Jumpbox interface. The page has a header with a search bar and navigation icons. Below the header, there's a breadcrumb trail: 'Home > Dashboards > DeepFlow Templates'. The main title is 'DeepFlow Templates' with an edit icon. Below the title, it says 'Manage folder dashboards and permissions'. There are two buttons: 'Folder actions' and 'New'. Below this, there are three tabs: 'Dashboards' (selected), 'Panels', and 'Alert rules'. A search bar for dashboards and folders is present. Below the search bar, there's a 'Filter by tag' dropdown and a 'Starred' checkbox. To the right, there are icons for a folder and a list, and a 'Sort' dropdown. The main content area is a table with columns 'Name' and 'Tags'. The table lists 20 templates, each with a checkbox and a 'DeepFlow' icon.

☐	Name	Tags
☐	Application - Service List	
☐	Application - Cloud Host	
☐	Application - Cloud Host Map	
☐	Application - DNS Monitoring	
☐	Application - Dubbo Monitoring - K8S	
☐	Application - K8s Ingress	
☐	Application - K8s Pod	
☐	Application - K8s Pod Map	
☐	Application - Redis Monitoring - Cloud	
☐	Application - Redis Monitoring - K8S	
☐	Application - Request Log	
☐	Application - Request Log - Cloud	
☐	Application - SQL Monitoring - Cloud	
☐	Application - SQL Monitoring - K8S	
☐	Application - 开箱即用	
☐	Continuous Profiling	

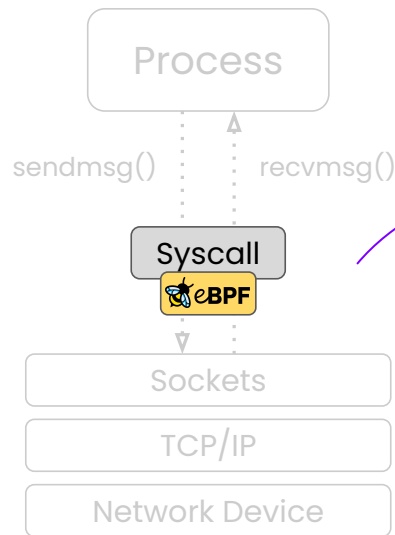
Reference(s):

- [Install Command](#)

Next Step

Open Source World

Next Step (Cont.)



OpenTelemetry



DeepFlow



Alloy



Pixie



coroot



CLOUD NATIVE
LANDSCAPE

Others

Jumpbox®

ทุกเทคโนโลยีเกิดขึ้นมาเพื่อแก้ไขปัญหา
แล้วปัญหาของเราตรงกับเทคโนโลยีนั้น
เราจึงหยิบสิ่งเหล่านั้นมาใช้

LGTM Ecosystem (Cont.)



Loki



Grafana



Tempo



Mimir

LGTM Ecosystem (Cont.)



Loki



Grafana



Tempo

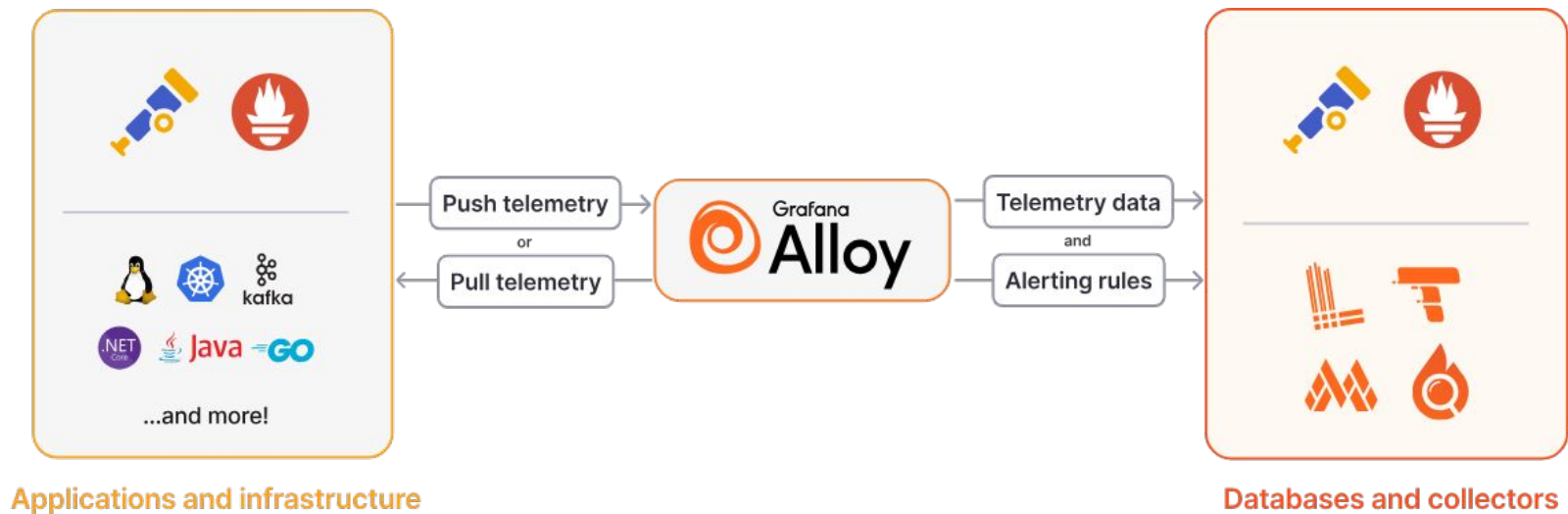


Mimir



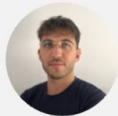
Alloy

LGTM Ecosystem (Cont.)



Actually useful AI with Grafana Cloud

Learn how Grafana Cloud's agentic AI systems can help you write queries, run investigations, manage dashboards, and resolve incidents all using natural language. Buffet lunch is provided after the session.



Filippo de Martinis
Senior Solutions Engineer

Scan to register →



Tuesday, 10 March
11:00 am - 1:00 pm



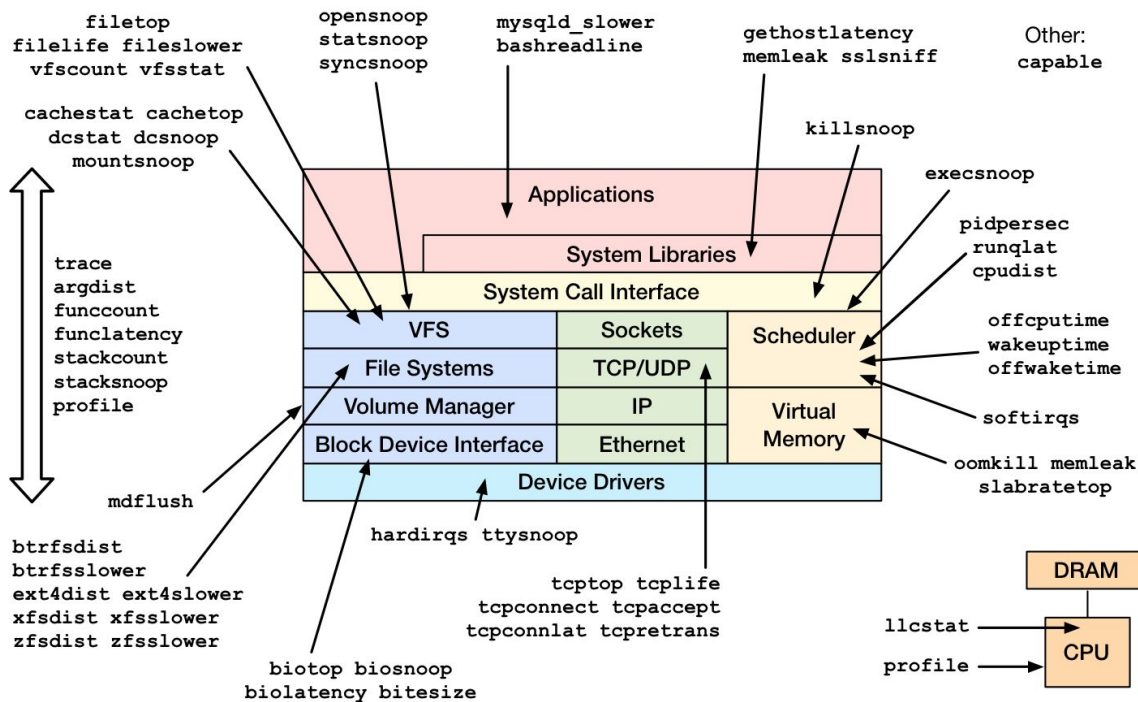
True Digital Park WEST
2nd Floor, Classroom 1



CLOUD NATIVE
BANGKOK
COMMUNITY GROUP

Back to Fundamental

Linux bcc/BPF Tracing Tools

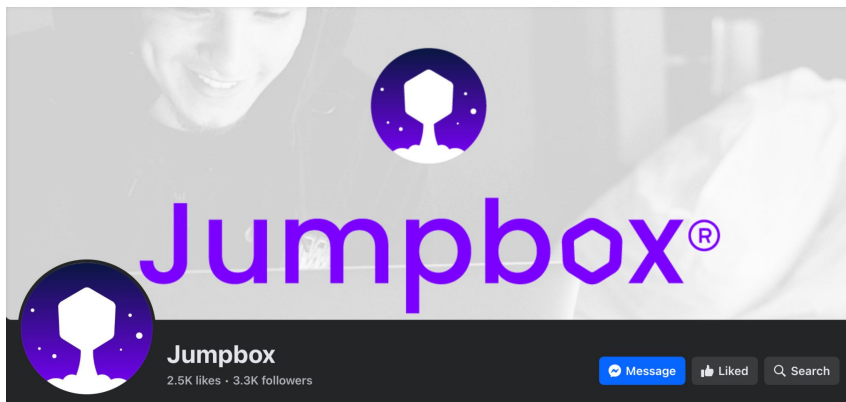


Reference(s):

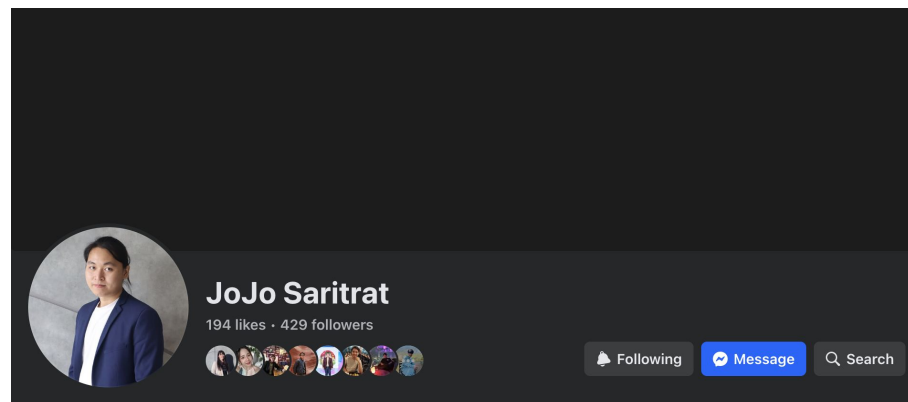
- [Linux BCC](#)
- [Brendan Gregg](#)

<https://github.com/iovisor/bcc#tools> 2016

facebook



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